CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 18

Name (Print):_____ Section:_____

1. Express Cost in terms of BigO for ArrayList:

Operation	Cost
get(index) add(index, element) remove(index) size()	

2. A sorting algorithm operates by _____ the list, _____ elements, and _____ them _____

3. Selection Sort has 2 parts: 1) _____ part at the beginning 2) _____ part is the rest of array

4. Suppose the selection sort algorithm from class is applied to the initial array:

9	5	7	10	18	1	12	8	16	4

In the boxes below, show the state of the array immediately following each of the first two iterations. Clearly mark the sorted part (with a vertical line) and the unsorted part of the array after each iteration for 1st, 2nd iterations. (Initially, the sorted part of the array is empty.)

0th iteration:

9	5	7	10	18	1	12	8	16	4

1st iteration:

2nd iteration:

5. Comparison Math Formula for selection algorithm:

_____

6. Summary Table for Selection Sort:

Case	BigO
Worst Best	

7. Suppose the insertion sort algorithm is applied to the initial array above. Show the state of the array immediately following each of the first three iterations of the outer loop. Clearly mark (as in 0th is alreday marked for you) the sorted part of the array after each iteration.

0st iteration:

9	5	7	10	18	1	12	8	16	4

1st iteration:

2nd iteration:

3rd iteration:

8. Summary Table for InsertionSort:

Case	BigO
Worst (reversed) Average Best (sorted)	

9. True/False Binary search must be sorted

10. Summary Table for Binary Search:

Case	BigO
Worst Best	

11. Summary Table for MergeSort:

Case	BigO
Worst Best	

12. Suppose the merge sort algorithm from class is applied to the initial array.

9	5	7	10	18	1	12	8	16	4

Show the state of the two sub-arrays immediately before the final merge.

